

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've spent 10 years building backend systems and developer platforms, with the last 3 focused on building tools and APIs that make engineering teams more effective. I architected developer platforms that reduced environment provisioning by 60x, designed REST APIs serving 5 teams and external partners, and built infrastructure that enables rapid iteration. I work in TypeScript, Python, and Java, building scalable APIs, microservices, and developer tooling. I'm drawn to problems at the intersection of infrastructure, developer experience, and system design. I lead through technical vision and mentorship, not authority.

Work Experience

Alberta Motor Association

Edmonton, Alberta

SOFTWARE ARCHITECT

Jan 2023 - Present

- Built self-service developer platform solo that reduced environment provisioning from 2-3 weeks to 2 hours (60x improvement). Adopted by 15 engineers across 3 teams, created 120+ environments. Unblocked major initiatives by enabling rapid iteration and production debugging. (Python, Terraform, AWS, Jenkins, Docker)
- Own API architecture for platform serving 5 internal teams and external partners. Designed API contracts, approve endpoint additions, maintain OpenAPI standards, and ensure consistency across services. APIs process \$4.3M in annual transactions. (Python, REST, OpenAPI, TypeScript)
- Architected technical infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, \$4.3M annual revenue, 2.8M events processed, 99.9% uptime. Won 2024 Guidewire Innovation Award. (Python, AWS Lambda, Terraform, TypeScript)
- Led modernization of document generation system from legacy C# to containerized microservices on EKS. Led 2-3 engineers. Integrated Datadog for observability (failure rates, success metrics, performance). Reduced environment provisioning from weeks to minutes. (Kubernetes, EKS, Docker, Datadog, Helm, C#)
- Set technical standards and provide architectural guidance for 12-person team through code reviews, RFC authorship, and technical documentation. Member of Architecture Steering Committee. Mentor engineers on system design, debugging, and best practices.

Arterys Inc. (Acquired by Tempus Labs)

Calgary, Alberta (Remote)

SOFTWARE DEVELOPER

Aug 2021 - Aug 2022

- Built backend services running medical AI algorithms, optimizing for low latency and high throughput. (TypeScript, Node.js, AWS)
- Improved algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, identifying and resolving issues before customer impact.

Alberta Motor Association

Edmonton, Alberta

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework from scratch that reduced defects by 30% on subsequent releases. Framework adopted org-wide, still in use 4+ years later across all development teams. Significantly improved developer velocity and release confidence. (Java/Gosu, JUnit)
- Led 5-6 person development team using TDD methodology. Delivered projects on time with no critical defects post-launch. Conducted code reviews, mentored junior developers, and collaborated with stakeholders on requirements and priorities.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local dev environments, reducing onboarding friction for new developers.

Pleasant Solutions

Edmonton, Alberta

SOFTWARE DEVELOPER

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD. (C#)

Lockheed Martin Canada

Halifax, Nova Scotia

SOFTWARE ENGINEER ASSOCIATE

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining real-time data streams from multiple sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system for capturing and replaying test scenarios, improving debugging and test coverage.

The King's University

Edmonton, Alberta

SOFTWARE DEVELOPER/RESEARCH ASSISTANT

May 2014 - Sep 2016

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.

Education

The Open University

MSC. COMPUTING (SOFTWARE ENGINEERING)

Currently pursuing Master's in Computing at The Open University.

Milton Keynes, England (Remote)

May 2024 - Ongoing

The King's University

B.S. IN COMPUTER SCIENCE

Edmonton, Alberta

Sep 2012 - Dec 2016

Skills

Developer Experience & Tooling	Built developer platforms (60x faster provisioning), self-service infrastructure tools, and test frameworks (30% defect reduction). Focus on removing friction and enabling teams to ship faster with better quality.
API Design & Architecture	Design and maintain REST APIs serving 5 teams and external partners, processing \$4.3M in annual transactions. Own API contracts, approve endpoint additions, and set design standards across organization.
Technical Leadership & Mentorship	Lead through technical vision, code reviews, and RFC authorship. Mentor engineers on debugging, testing, and architectural decisions. Member of Architecture Steering Committee. Advocate for engineering excellence and pragmatic approaches.
Scalable Backend Systems	Built systems with 99.9% uptime serving multiple teams. Experience with microservices, containerization (Docker, Kubernetes), cloud infrastructure (AWS), and observability (Datadog). Strong focus on performance and reliability.

Technologies

Languages	TypeScript, Python, Java, Bash, C#
Backend & APIs	REST APIs, OpenAPI, Microservices, CI/CD, Event-Driven Architecture
Databases	PostgreSQL, Aurora/RDS, Redis
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM), Kubernetes, Docker, Terraform, Helm
DevOps & Tooling	Test Automation (JUnit, Playwright), Datadog, Elasticsearch, Kibana